PROJECT TITLE

"Design and Implementation of a System for the Automation of the Structural Design of Exterior Works using Neural Networks and Optimization Algorithms: A Design Science Approach for the Efficiency of Construction Documents (CDs) in Rural Schools of Peru"

1. PROBLEM STATEMENT

1.1 The Macro Context: Chronic Inefficiency in Rural Educational Infrastructure

In Peru, the gap in educational infrastructure in rural areas constitutes one of the most critical challenges for socioeconomic development and social equity. Although the Peruvian State allocates significant budgets through mining canon transfers and direct funds to regional and local governments, there is an alarming contradiction between the availability of financial resources and the physical execution of public construction projects (Chang & Villegas, 2024). This problem manifests itself mainly through two phenomena: chronic budget under-execution and the proliferation of stalled or abandoned construction projects. An emblematic case occurs in the district of San Marcos (Áncash), the municipality with the highest revenue collection in the country, which left more than 500 million soles unexecuted in a single fiscal year while thousands of families lack basic services and road connectivity (Chang & Villegas, 2024).

The underlying causes of this problem are not a scarcity of funds, but rather technical inefficiency during project formulation. In fact, macroeconomic estimates based on World Bank data show that nearly 45% of the public projects started in Peru between 2012 and 2023 were completely abandoned or suffered critical stalling due to deficiencies in management and investment planning (Chang & Villegas, 2024).

1.2 The Technical Bottleneck: The Formulation of Construction Documents (CDs)

Under the framework of the National System of Multiannual Programming and Investment Management (Invierte.pe), the physical start of any school infrastructure work requires the mandatory approval of the Construction Documents (CDs). This package is a highly complex engineering study that normatively includes descriptive specifications, quantity take-offs (metrados), budgets, and structural design.

In rural public schools, the so-called exterior works (perimeter fences, retaining walls due to rugged topography, accessibility ramps, and sports courts) represent a critical portion of the project, indispensable to guarantee the functionality of a school. However, the current process of structural design for these exterior works is carried out using traditional methodologies that rely entirely on manual or semi-assisted calculations by civil engineers. This approach presents three severe limitations:

- **High processing times:** The modeling and calculation of moments take weeks for each exterior work module.
- **Lack of optimization:** Due to administrative pressures, conservative over-sizing is usually adopted, which unnecessarily increases project costs, or under-sizing is applied, generating additional costs during execution.
- **Technical inconsistencies:** Human errors generate recurring observations by State evaluators such as PRONIED (Peruvian National Program of Investments in Education), causing endless loops of correction that delay the project by months or years.

1.3 The Knowledge Gap in Artificial Intelligence and Design Science

From the perspective of computer science, structural design is a multi-objective and high-dimensional optimization problem, which seeks to minimize costs and times while ensuring strict compliance with the National Building Code of Peru (Standards E.020, E.030, E.060, and E.090). Despite global advances in Artificial Intelligence (AI), the application of generative models, artificial neural networks (ANN), and optimization algorithms has been concentrated almost exclusively on commercial mega-structures or skyscrapers within the contexts of developed countries. There is a gap in knowledge and technological innovation regarding the development of AI tools adapted for social public infrastructure engineering in rural Latin American contexts. Currently, there is no validated technological artifact (application/software system) under Design Science that integrates neural networks with optimization algorithms to automate the structural design workflow of exterior works for rural public schools. The result of this technological gap is the perpetuation of excessively long formulation periods and Construction Documents (CDs) with design errors that lead directly to stalled construction projects in the field described by Peruvian national economic literature (Chang & Villegas, 2024).

2. PARADIGM STATEMENT

This research formally adopts the Design Science Research paradigm with a three-cycle iteration design (relevance, design, and rigor).

- **Rejection of the positivist approach:** A purely positivist approach is rejected because statistically measuring the historical inefficiency of spending in Peru does not resolve the project delays, nor does it generate by itself the automated artifact that civil engineers need in the field to accelerate technical formulation.
- **Rejection of the interpretivist approach:** A purely interpretive approach is rejected because, although qualitative interviews make visible the frustrations of project formulators, the understanding of the human problem is insufficient if it does not materialize into a technical, mathematical, and algorithmic solution capable of performing structural calculations autonomously.
- **Rejection of mixed methods:** The traditional mixed methods approach is rejected since the main goal of this doctoral thesis is not the sequencing or triangulation of quantitative and qualitative data to explain a social phenomenon, but the creation and empirical validation of a disruptive technology (the intelligent computer system) non-existent in the public sector.

The logic of the chosen paradigm lies in the fact that scientific truth and validity in this study are discovered and demonstrated through the utility, novelty, and rigor of the constructed artifact. In this case, the system based on neural networks and optimization does not seek to describe the reality of budget under-execution in the regions of Peru, but to transform it directly by drastically reducing engineering times. For this research, reality is composed of practical engineering problems and technological artifacts that interact with socio-technical systems. The inefficiency in the Construction Documents (CDs) of rural schools in Peru and the limitations of traditional design methods are real flaws of the operational environment. The artificial intelligence system to be developed will become part of that reality as a concrete, measurable, and evaluable object based on its computational and structural performance. Reliable knowledge is acquired through the construction, evaluation, and iteration of functional prototypes. We will not limit ourselves to neutral observation; instead, we will intervene in the problem to solve it. The system will function and generate valid scientific knowledge when subjected to algorithmic stress tests, load validation according to the Peruvian technical building standards (E.020/E.030/E.060/E.090), and rigorous time comparisons against traditional human design.

A structured methodological process is followed in three major phases:

- **Construction:** Development of artificial neural network architectures and optimization algorithms for the design of exterior works (retaining walls, perimeter fences, sports courts).
- **Evaluation:** Performance testing of automated designs using actual records from previous rural school projects.
- **Iteration:** Refinement of the code based on structural error margins and processing time optimization until reaching the technical viability standards required in Peru.

The core values of this research are practical utility, technical rigor, and social impact. Priority is given to the development of a tool that optimizes State resources to reduce the preparation times of Construction Documents (CDs) in the specialty of Structures and mitigate the stalling and abandonment of construction projects in the most vulnerable areas of the country (rural schools). The ethical commitment of this research is manifested in the delivery of a transparent, reproducible, and computationally exact artifact, whose ultimate goal is to accelerate access to dignified educational infrastructure in Peru.